

ES03 Aleph Drone Expansion Board Data Sheet

April 2025
• Carrier board revision: **2.0.0**

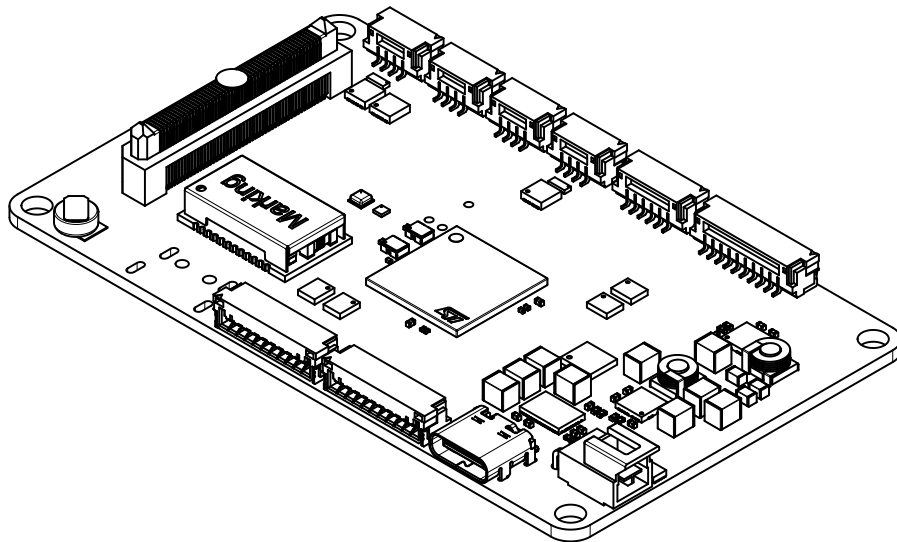


Figure 1: Aleph Drone Expansion Board

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Overview

The Aleph Drone Expansion Board is a versatile, standalone flight controller built around the high-performance STM32H7 microcontroller. Designed for seamless integration into drone and robotics systems, it features a flight sensor suite, exposed communication peripherals, and an onboard debugger/flasher accessible via a single USB-C connection—streamlining development and deployment. This open development platform supports popular autopilot firmware such as PX4 and Betaflight, as well as Elodin's own Roci control stack or any custom flight software. For compute-intensive missions, it can interface directly with the Aleph Carrier Board, enabling tight coupling between low-level control and high-level autonomy.

Key Features

- Peripheral support: 3x UART, 2x CAN, I2C, 16x PWM, 8x GPIO, MircoSD, USB FS (12 Mbps) with PD
- Flight sensor suite: 2x IMU (with communication redundancy), Barometer, Magnetometer
- Visual status indicators: 7x status LEDs

Specifications

Environmental

Minimum operating temperature (°C)	-25
Maximum operating temperature (°C)	80
ROHS compliant	Yes
Vibration qualification	Please contact us for further information

Mechanical

Dimensions (mm)	Refer to Figure 2, 3, 4
Base Material	Isola 370HR
Surface finish	ENIG

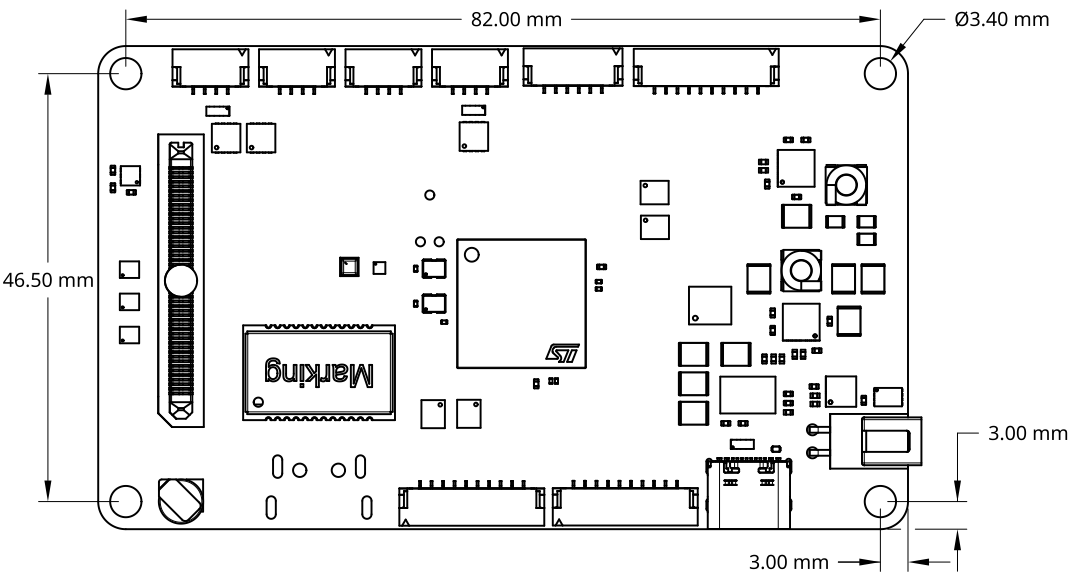


Figure 2: Aleph Drone Expansion Board Top View

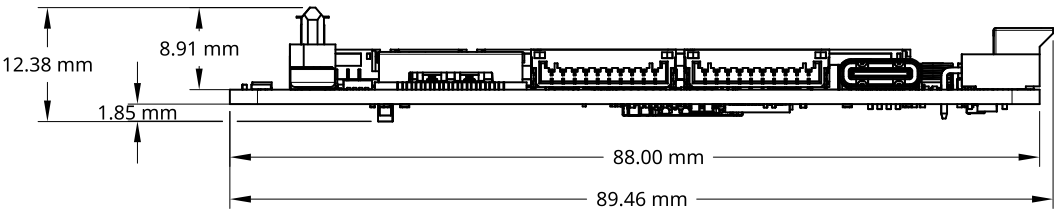


Figure 3: Aleph Drone Expansion Board Front View



Figure 4: Aleph Drone Expansion Board Side View

Electrical and Hardware

Minimum operating supply voltage (V)	5
Maximum operating supply voltage (V)	28
Standard power consumption (W)	3
Maximum peripheral power supply (W)	15
Peripheral VDD (V)	5
Peripheral logic level (V)	3.3
Peripheral VDD rail current monitoring	Yes
Switchable Peripheral VDD rail	Yes
Over-current protection	Yes
Power input reverse current protection	Yes
ESD protection	Yes
VBAT-IN voltage monitoring	Yes
Regulated internal voltage monitoring	Yes
FRAM Capacity (Kbit)	16

Sensors

IMU	BMI323 or BMI270
Barometer	BMP581
Magnetometer	BMM350

Microcontroller Specs

The Aleph Drone Expansion Board has 2 MCUs, an STM32H7 as the main micro controller and the an RP2040 as the debugger and flasher. This allows the RP2040 to serve several useful functions on the FC board. When flashed with [debugprobe](#) firmware, it can serve as a CMSIS-DAP probe and USB-to-UART bridge. This provides a convenient way to flash new firmware onto the STM32 over USB using standard tools such as [OpenOCD](#) or [probe-rs](#). Since the RP2040 can monitor the STM32's UART1 and pull its nRST low, it can also function as a watchdog.

NOTE: the RP2040 is pre-flashed with debugprobe firmware.

Cores	• ARM Cortex-M7 @ 480MHz • ARM Cortex-M4 @ 240MHz
Flash memory capacity (MB)	2
RAM (MB)	1

Connectivity

Board to Board

Software

Autopilot Support

The Aleph Drone Expansion Board ships with an image of the Betaflight autopilot. It is also compatible with PX4. The system can also be flashed with custom flight control software or Elodin's own Roci control stack or any custom flight

software. For instruction on how to flash the system, refer to the [Elodin Docs page](#).